



Academic year 2024-2025 (Even Sem)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Date	5 th June 2025	Maximum Marks	(10+50) Marks
Course Code	CD363IA	Duration	(30+90) = 120 Min
Sem	VI Sem	CIE-2	
UG/PG	UG	Faculty: AS	
Data Analytics & Visualization(CD363IA)			
Scheme & Solution			

Q. No.	QUIZ	M															
1.	Describe the mechanisms for Aggregation in Crowdsourcing Sol: Mechanisms for Aggregation in Crowdsourcing refer to methods used to combine multiple inputs, responses, or judgments from a crowd to arrive at a reliable or accurate outcome. Since crowdsourcing involves collecting data or solutions from a large group of contributors (often with varying expertise), effective aggregation is essential to ensure quality and consistency. Collecting wisdom from a set of responses requires using the right aggregation mechanism. For estimating numerical quantities, standard techniques like plotting the frequency distribution and computing summary statistics are appropriate. - Both the mean and median implicitly assume that the errors are symmetrically distributed. A quick look at the shape of the distribution can generally confirm or reject that hypothesis. - The median is, more appropriate choice than the mean in such aggregation problems. It reduces the influence of outliers, which is a particular problem in the case of mass experiments. Removing outliers is a very good strategy, but we may have other grounds to judge the reliability of our subjects, such as their performance on other tests where we do know the answer. Taking a weighted average, where we give more weight to the scores deemed more reliable, provides a way to take such confidence measures into account. For classification problems, voting is the basic aggregation mechanism.	2															
2.	State the principles underlying design and developing a Visualization Aesthetic. Sol: The principles underlying design and developing a Visualization Aesthetics are: - Maximize data-ink ratio - Minimize the lie factor - Minimize chartjunk - Use proper scales and clear labelling - Make effective use of color - Exploit the power of repetition	2															
3.	Identify the Chart type for the following use cases: - relationship between two numeric variables. - to represent part-to-whole relationships - geographic or spatial data - The frequency distribution of a continuous variable <table border="1"> <thead> <tr> <th>Sl. No.</th><th>Use Case</th><th>Appropriate Chart Type</th></tr> </thead> <tbody> <tr> <td>a)</td><td>To show relationship between two numeric variables</td><td>Scatter Plot</td></tr> <tr> <td>b)</td><td>To represent part-to-whole relationships</td><td>Pie Chart</td></tr> <tr> <td>c)</td><td>To visualize geographic or spatial data</td><td>Data Map / Choropleth Map</td></tr> <tr> <td>d)</td><td>To display frequency distribution of a continuous variable</td><td>Histogram</td></tr> </tbody> </table>	Sl. No.	Use Case	Appropriate Chart Type	a)	To show relationship between two numeric variables	Scatter Plot	b)	To represent part-to-whole relationships	Pie Chart	c)	To visualize geographic or spatial data	Data Map / Choropleth Map	d)	To display frequency distribution of a continuous variable	Histogram	2
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4.	List any two good applications of Turkers and any two bad uses of crowdsourcing. Good applications of Turkers include: (List Any 2)	2															

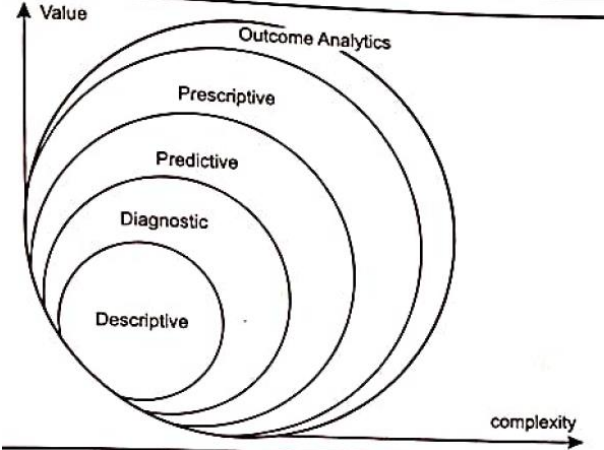


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	- Measuring aspects of human perception - Obtaining training data for machine learning classifiers: - Obtaining evaluation data for computer systems - Putting humans into the machine - Independent creative efforts - Economic/psychological experiments: Bad uses of Crowdsourcing include: (List Any 2) - Any task that requires advanced training - Any task you cannot specify clearly: - Any task where you cannot verify whether they are doing a good job - Any illegal task, or one too inhuman to subject people to	
5.	Write the pictorial representation indicating the Coverage of Applied statistics.	2
Q. No.	TEST	M
1.	Illustrate how gamification can enhance the effectiveness of crowdsourced data collection with any three successful examples. There is an alternative to paying people to annotate or transcribe your data. Instead, make things so much fun that people will work for you for free. Games with a purpose (GWAP) are systems which disguise data collection as a game people want to play, or a task people themselves want done. With the right combination of game, motive, and imagination, amazing things can be done. Successful examples include: CAPTCHAs for optical character recognition (OCR): CAPTCHAs are those distorted text images you frequently encounter when creating an account on the web. They demand that you type in the contents of text strings shown in the image to prove that you are a human, thus enabling them to deny access to bots and other programmed systems. - ReCAPTCHAs were invented to get useful data from the over 100 million CAPTCHAs displayed each day. Two text strings are displayed in each, one of which the system checks in order to grant entry. The other represents a hard case for an OCR system that is digitizing old books and newspapers. The answers are mapped back to improve the digitization of archival documents, transcribing over 40 million words per day. Psychological/IQ testing in games/apps: Psychologists have established five basic personality traits as important and reproducible aspects of personality. Academic	10



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	psychologists use multiple-choice personality tests to measure where individuals sit along personality scales for each of the big five traits: openness, conscientiousness, extroversion, agreeableness, and neuroticism. ○ By turning these surveys into game apps ("What are your personality traits?") psychologists have gathered personality measurements on over 75,000 different people, along with other data on preferences and behavior. This has created an enormous data set to study many interesting issues in the psychology of personality. ➤ The FoldIt game for predicting protein structures: Predicting the structures formed by protein molecules is one of the great computational challenges in science. Despite many years of work, what makes a protein fold into a particular shape is still not well understood. ○ FoldIt (https://fold.it) is a game challenging non-biologists to design protein molecules that fold into a particular shape. Players are scored as to how closely their design approaches the given target, with the highest scoring players ranked on a leader board. Several scientific papers have been published on the strength of the winning designs. • The key to success here is making a game that is playable enough to become popular. This is much harder than it may appear. There are millions of free apps in the app store, mostly games. Very few are ever tried by more than a few hundred people, which is nowhere near enough to be interesting from a data collection standpoint. Adding the extra constraint that the game generate interesting scientific data while being playable makes this task even harder. • Motivational techniques should be used to improve playability. Keeping score is an important part of any game, and the game should be designed so that performance increases rapidly at first, in order to hook the player. Progress bars provide encouragement to reach the next level. Awarding badges and providing leader boards seen by others encourages greater efforts. The primary design principle of games such as FoldIt is to abstract the domain technicality away, into the scoring function.	
2.	Discuss the five different types of big data analytics covering the relevance, key points, and suitable real-world examples of each. 	10



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Descriptive: What's happening in my business?

- Comprehensive, accurate and live data.
- Effective visualisation

Diagnostic: Why is it happening?

- Ability to drill down to the root-cause
- Ability to isolate all confounding information

Predictive: What's likely to happen?

- Business strategies have remained fairly consistent over time
- Historical patterns being used to predict specific outcomes using algorithms
- Decisions are automated using algorithms and technology

Prescriptive: What do I need to do?

- Recommended actions and strategies based on champion/challenger testing strategy outcomes.
- Applying advanced analytical techniques to make specific recommendations.

Descriptive Analytics

Key points:

- Backward looking
- Focused on descriptions and comparisons
- Pattern detection and descriptions
- MECE (mutually exclusive and collectively exhaustive) categorization
- Category development based on similarities and differences (segmentation)

Diagnostic Analytics

Key points:

- Backward looking
- Focused on causal relationships and sequences
- Relative ranking of dimensions/variable based on inferred explanatory power)
- Target/dependent variable with independent variables/dimensions
- Includes both frequentist and Bayesian causal inferential analyses

Predictive Analytics

Key points:

- Forward looking
- Focused on non-discrete predictions of future states, relationship, and patterns
- Description of prediction result set probability distributions and likelihoods
- Model application
- Non-discrete forecasting (forecasts communicated in probability distributions)

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Prescriptive Analytics

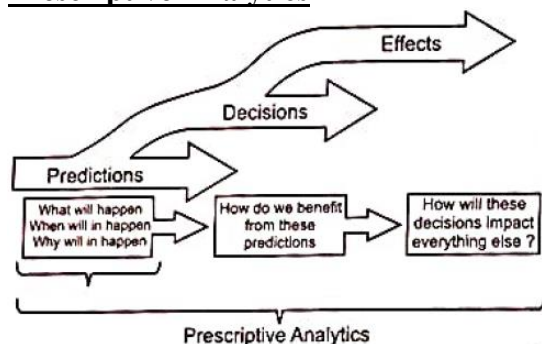


Fig. 3.3: Development of Prescriptive Analytics

Key points:

- Forward looking
- Focused on optimal decisions for future situations
- Simple rules to complex models that are applied on an automated or programmatic basis
- Discrete prediction of individual data set members based on similarities and differences
- Optimization and decision rules for future events

Outcome Analytics:

Key points:

1. Backward looking, Real-time and Forward looking
2. Focused on consumption patterns and associated business outcomes
3. Description of usage thresholds
4. Model application

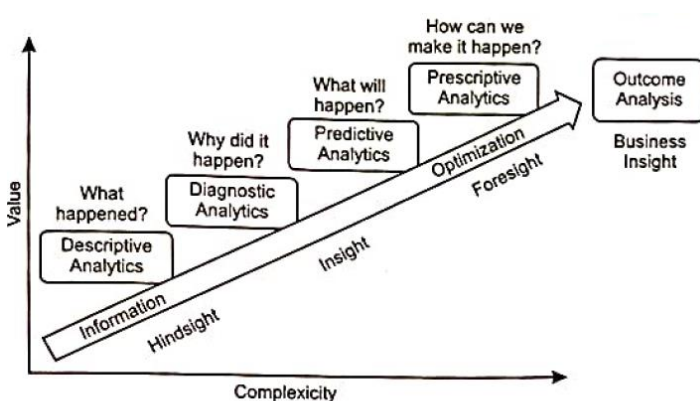


Fig. 3.4: Optimization of business data to Create Business Insights

3a.

“A right plot for a given data set needs to be produced, not just the first thing that comes to mind.” In this context, write the complete pictorial representation of a handy decision tree to help identify the best visual representation for representing data (from Abela).

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Solution :

(a) Relative Frequencies:

Total customers = 12 + 20 + 30 + 18 + 10 = 90

Time Interval	Frequency	Relative Frequency = Frequency / Total
08:00–10:00	12	12 / 90 = 0.133
10:00–12:00	20	20 / 90 = 0.222
12:00–14:00	30	30 / 90 = 0.333
14:00–16:00	18	18 / 90 = 0.200
16:00–18:00	10	10 / 90 = 0.111

----- 3.5 M

(b) Probability (12:00–14:00) = Relative Frequency = 30 / 90 = 0.333

----- 1.5 M

4.

With suitable use cases and mathematical formulations illustrate the principles of Maximizing Data-Ink Ratio and Minimizing the Chartjunk in Visualization. Also support your answer with relevant visualization graphs.

Maximizing Data-Ink Ratio

----- 5M

In any graphic, some of the ink is used to represent the actual underlying data, while the rest is employed on graphic effects. Visualizations should focus on showing the data itself. We define the data-ink ratio to be:

$$\text{Data-Ink Ratio} = \frac{\text{Data-Ink}}{\text{Total ink used in graphic}}$$

Figure 6.5 presents average salary by gender (Bureau of Labor Statistics, 2015), and helps clarify this notion.

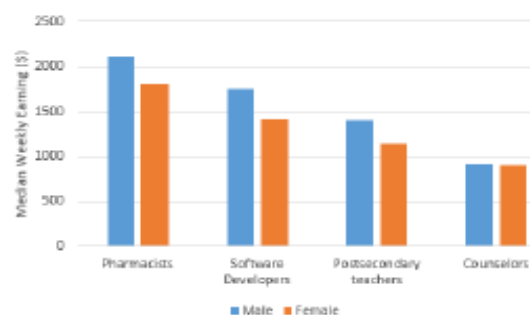
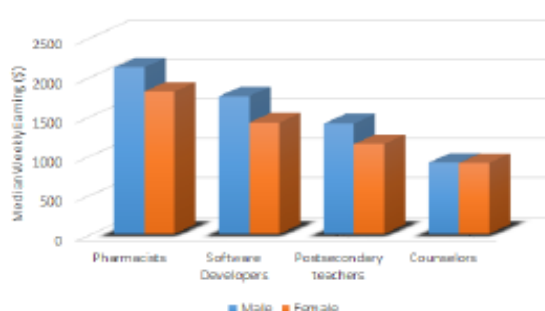


Figure 6.5: Three-dimensional monoliths casting rendered shadows (l) may look impressive. But they really are just chartjunk, which serves to reduce the clarity and data-ink ratio from just showing the data (r).

- Maximizing the data-ink ratio lets the data talk, which is the entire point of the visualization exercise in the first place. The flat perspective on the right permits a fairer comparison of the heights of the bars. The colors do a nice job enabling us to compare.
- There are more extreme ways to increase the data-ink ratio. Instead of bars, the same information could be conveyed by plotting a point of the appropriate height, and would clearly be an improvement were we plotting much more than the eight points shown here. Be aware that less can be more in visualizing data.

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Minimizing Chartjunk

----- 5M

Extraneous visual elements distract from the message the data is trying to tell. In an exciting graphic, the data tells the story, not the chartjunk.

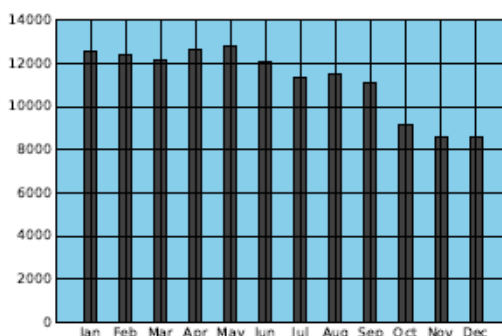


Figure 6.7: A monthly time series of sales. How can we improve/simplify this bar chart time series?

Figure 6.7 presents a monthly time series of sales at a company beginning to encounter bad times. The graphic in question is a bar plot, a perfectly sound way to represent time series data, and is drawn using conventional, perhaps default, options using a reasonable plotting package.

We can simplify this plot by removing elements to make the data stand out better as in Figure 6.8, which presents a series of four successive simplifications to this chart. The critical operations are:

- Jailbreak your data (upper left):
- Stop throwing shade (upper right):
- Think outside the box (lower left):
- Make missing ink work for you (lower right):

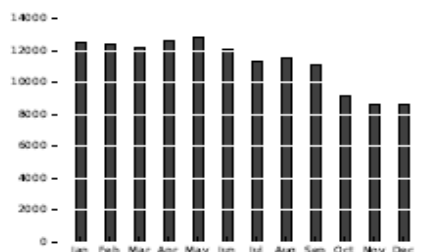
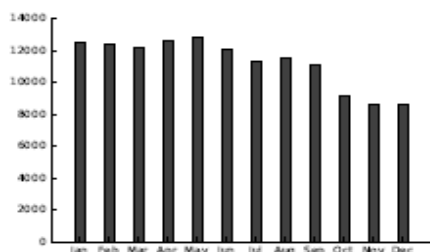
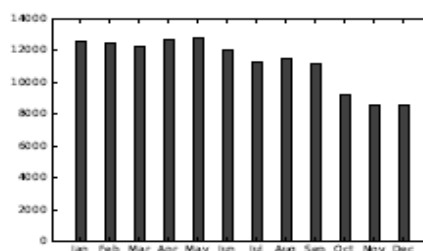
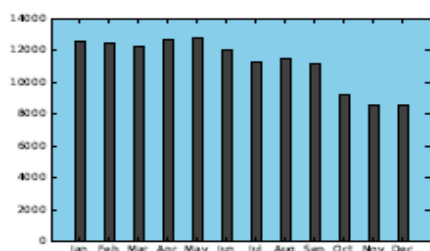


Figure 6.8: Four successive simplifications of Figure 6.7, by removing extraneous non-data elements.

The architect Mies van der Rohe famously said that "less is more." Removing elements from plots often improves them far more than adding things.

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5a.

Illustrate the characteristics of Great Visualizations considering the two case studies

i) Snow's Cholera Map ii) New York's Weather Year.

Snow's Cholera Map

A particularly famous data map changed the course of medical history. Cholera was a terrible disease which killed large numbers of people in 19th-century cities. The plague would come suddenly and strike people dead, with the cause a mystery to the science of the day.

John Snow plotted cholera cases from an epidemic of 1854 on a street map of London, hoping to see a pattern. Each dot in Figure 6.28 represented a household struck with the disease.



Figure 5(a) – (i)Cholera deaths center around a pump on Broad street, revealing the source of the epidemic.

Snow noticed a cluster of the cases centered on Broad Street. Further, at the center of the cluster was a cross, denoting a well where the residents got their drinking water. The source of the epidemic was traced to the handle of a single water pump. They changed the handle, and suddenly people stopped getting sick. This proved that cholera was an infectious disease caused by contaminated water, and pointed the way towards preventing it.

New York's Weather Year

It is almost worth enduring a New York winter to see a particular graphic which appears in The New York Times each January, summarizing the weather of the previous year.

Figure 6.29 presents an independent rendition of the same data, which captures why this chart is exciting. For every day of the year, we see the high and low temperature plotted on a graph, along with historical data to put it in context: the average daily high and low, as well as the highest/lowest temperatures ever recorded for that date.

First, it shows $6 \times 365 = 2190$ numbers in a coherent manner, which facilitates comparisons on the sine curve of the seasons:

- We can tell when there were hot and cold spells, and how long they lasted.
- We can tell what days had big swings in temperature, and when the thermometer hardly moved at all.

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- We can tell whether the weather was unusual this year. Was it an unusually hot or cold year, or both? When were record high/low set, and when did they get close to setting them?

This single graphic is rich, clear, and informative.

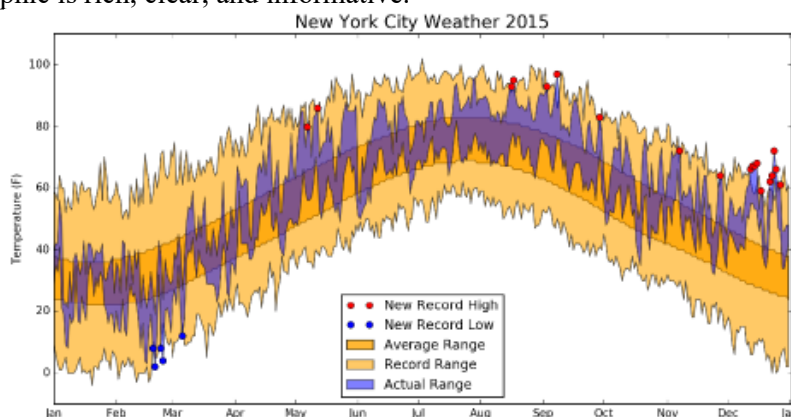


Figure 5(a) – (ii): This weather year in review displays a clear story in over 2000 numbers.

5b.

Discuss the best practices for bar plots and pie charts with relevant use cases and visualization graphs.

Bar plots and pie charts are tools for presenting the relative proportions of categorical variables. Both work by partitioning a geometric whole, be it a bar or a circle, into areas proportional to the frequency of each group. Both elements are effective in multiples, to enable comparisons. Indeed, partitioning each bar into pieces yields the stacked bar chart.

Figure 5b-i) shows voter data from three years of U.S. presidential elections, presented as both pie and bar charts. The blue represents Democratic votes, the red Republican votes. The pies more clearly shows which side won each elections, but the bars show the Republican vote totals have stayed fairly constant while the Democrats were generally growing. Observe that these bars can be easily compared because they are left-justified.

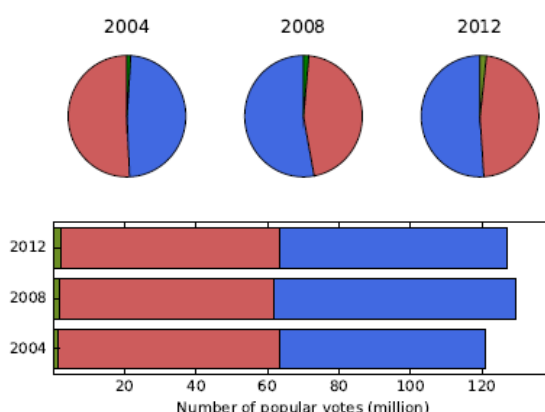


Figure 5b-i) : Voter data from three U.S. presidential elections. Bar plots and pie charts display the frequency of proportion of categorical variables. Relative magnitudes in a time series can be displayed by modulating the area of the line or circle.

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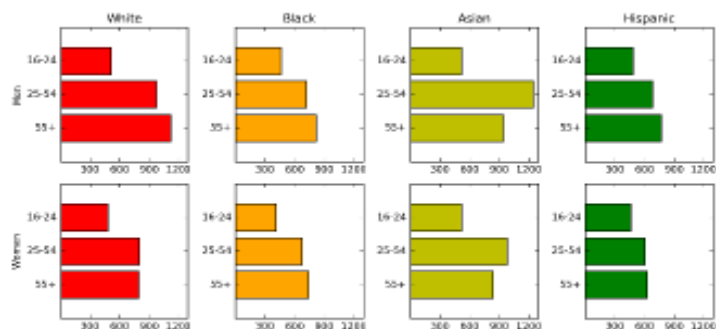


Fig 5b-ii) Small multiple bar plots/tables are excellent ways to represent multivariate data for comparison.

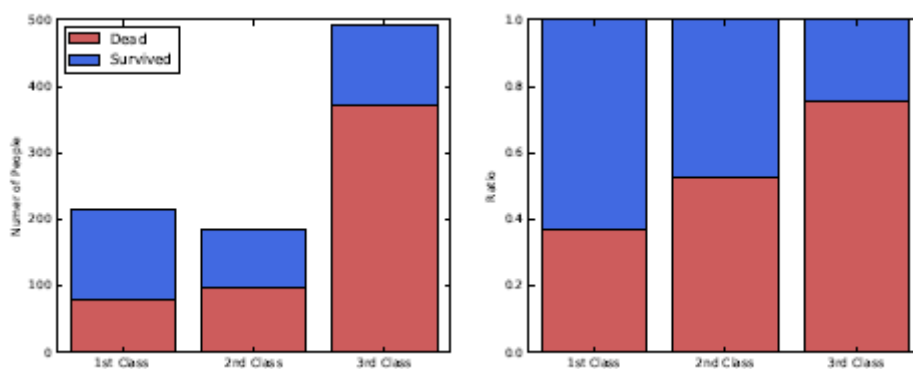


Fig 5b-iii) Stacked bar charts illustrating the survivorship rate on the doomed ship Titanic, by ticket class. The histogram (left) informs us of the size of each class, but scaled bars (right) better capture proportions. Primary conclusion: you were better off not to be traveling steerage (third class).

Best practices for bar plots and pie charts include:

- Directly label slices of the pie
- Use bar charts to enable precise comparisons
- Scale appropriately, depending upon whether you seek to highlight absolute magnitude or proportion